Mini Project 2

Autism Screening Survey

Using a supervised classification model to optimise survey performance.

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01

Overview

A high-level summary & purpose.

Overview

Context

We are reviewing a dataset on an Autistic Spectrum Disorder Screening for Adults.

This is a survey with 10 questions that a respondent needs to fill out
to **screen** whether they require a further specialist diagnostic assessment to check if they are on the spectrum. There are additional demographic and clinical background questions too.

The **outcome** of the screening is provided and is denoted as the output column yes/no for further assessment.

Question

Can we reduce the number of questions to achieve a comparable predictive performance to the original questionnaire?

Business Case

Can we reduce the number of questions thereby decreasing likelihood of drop-offs, poor data, and increase completions of survey.

Success Metrics

Recall Rate and Accuracy

02

Data Background

A review of the data.

Data Background

Data Available

703
Records

14
Features

Features	Data Type	Data Category
A2_Score	Boolean	Screening Questions for survey
A3_Score	Boolean	
A4_Score	Boolean	
A5_Score	Boolean	
A7_Score	Boolean	
A8_Score	Boolean	
A9_Score	Boolean	
age	Continuous	Demographic Questions
gender	Categorical	
ethnicity	Categorical	
jaundice	Boolean	Clinical Background Questions
family_pdd	Boolean	
used_app_before	Boolean	
relation	Categorical	

The Survey

Survey Preview

- Respondents have 4 options to choose.
- Agree and Disagree responses are grouped together.
- 1, 7, 8, & 10 gives 1 point for **agree**
- 2-6 & 9 receives 1 point for **disagree**
- Sum the total points. if respondent scores 6 or higher refer to further assessment.

Please tick one option per question only:		**Definitely Agree**	**Slightly Agree**	**Slightly Disagree**	**Definitely Disagree**
1	I often notice small sounds when others do not				
2	I usually concentrate more on the whole picture, rather than the small details				
3	I find it easy to do more than one thing at once				
4	If there is an interruption, I can switch back to what I was doing very quickly				
5	I find it easy to 'read between the lines' when someone is talking to me				
6	I know how to tell if someone listening to me is getting bored				
7	When I'm reading a story I find it difficult to work out the characters' intentions				
8	I like to collect information about categories of things (e.g. types of car, types of bird, types of train, types of plant etc)				
9	I find it easy to work out what someone is thinking or feeling just by looking at their face				
10	I find it difficult to work out people's intentions				

SCORING: Only 1 point can be scored for each question. *Score 1 point for Definitely or Slightly Agree on each of items 1, 7, 8, and 10. Score 1 point for Definitely or Slightly Disagree on each of items 2, 3, 4, 5, 6, and 9.* If the individual scores **6 or above**, consider referring them for a specialist diagnostic assessment.

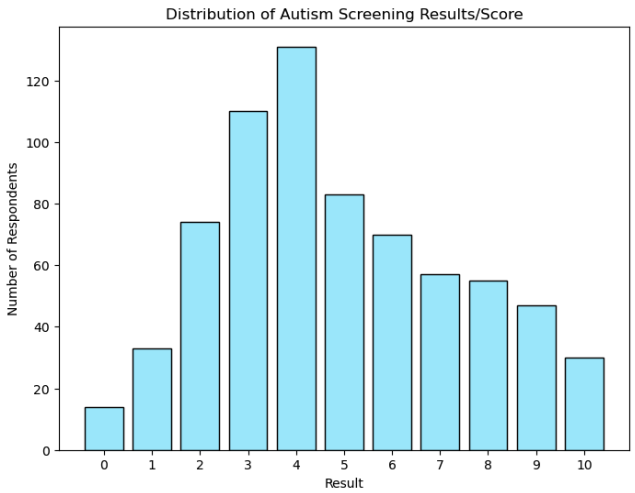
03

EDA

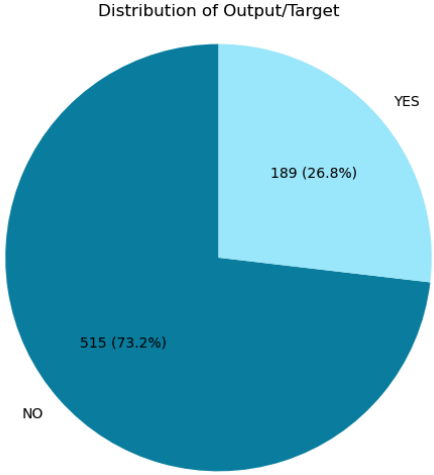
Exploring the data and the cleaning process.

EDA

Results score and Output distribution



High frequency of result scores sitting between a score of 3 and 4.



Higher 'No' outputs at 73%

Other findings

- Result score of 7 or higher are considered a **yes** to further screening.
- Looked at the clinic background ~90% skewed to no
- Age range 17-64, mean 29

EDA

Missing Values

Features with missing vales:

- Age - 2 values
- Ethnicity – 95 values
- Relation – 95 values

A process was needed to impute this data for classification modelling.

- Age – Mean age of the associated country
- Ethnicity – The percentage share of ethnicity by country
- Relation – Relation type with highest frequency

ethnicity	'Middle Eastern '	'South Asian'	Asian	Black	Hispanic	Latino	Others	Pasifika	Turkish	White-European	others	NaN
country_of_res												
'Costa Rica'	0	0	0	0	0	1	0	0	0	0	0	0
'Czech Republic'	0	0	0	0	0	0	0	0	0	1	0	0
'Hong Kong'	0	0	0	0	0	0	0	0	0	0	0	1
'New Zealand'	7	9	26	3	0	0	5	6	0	19	0	5
'Saudi Arabia'	2	0	0	0	0	0	0	0	0	0	0	2
'Sierra Leone'	0	0	1	0	0	0	0	0	0	0	0	0
'South Africa'	0	0	0	0	0	0	0	0	0	2	0	0
'Sri Lanka'	0	1	13	0	0	0	0	0	0	0	0	0
'United Arab Emirates'	53	1	2	5	0	0	3	1	2	0	0	15
'United Kingdom'	2	0	2	4	0	1	2	0	0	65	0	1
'United States'	5	1	4	10	11	1	7	1	1	70	1	1

EDA

Data Cleaning

Outline of additional data cleaning & manipulation:

- Ethnicity
  - Merged 'Others'
  - Grouped Hispanic and Latino
  - Merged Turkish to central Asia
- Age
  - Dropped 383yr old
  - Converted to Integer
- Dropped columns
  - country_of_res, results, output, age_desc
- Converted to binary from string
  - jaundice, famly_pdd, used_app_before
- Variables dummied
  - gender, ethnicity, relation
  - Dropped one column for each due to multicollinearity

04

Modelling

A comparison of different models and their performance.

Modelling

Building the model

Process

- Chose to drop A_score: 1, 6 and 9
- We chose to proceed with three models
  - GaussianNB
  - Logistic Regression
  - SVM
- A standard scaler was applied to Logistic Regression and SVM due to continuity of age.
- Initial modelling was performed using sklearn default parameters.
- A GridSearchCV was performed to further optimise results.

Cross tab of Yes / No answer distribution for A scores

	0	1	Total
A1_Score	0.279	0.721	1.0
A2_Score	0.546	0.454	1.0
A3_Score	0.542	0.458	1.0
A4_Score	0.504	0.496	1.0
A5_Score	0.501	0.499	1.0
A6_Score	0.716	0.284	1.0
A7_Score	0.582	0.418	1.0
A8_Score	0.350	0.650	1.0
A9_Score	0.676	0.324	1.0
A10_Score	0.425	0.575	1.0

Modelling Comparison

93% Accuracy

92% Recall

Naive Bayes

Confusion Matrix	Predicted No Autism	Predicted Suspected Autism
Actual No Autism	TN 86	FP 6
Actual Suspect Autism	FN 4	TP 45

91% Accuracy

82% Recall

Logistic Regression

Confusion Matrix	Predicted No Autism	Predicted Suspected Autism
Actual No Autism	TN 89	FP 3
Actual Suspect Autism	FN 9	TP 40

91% Accuracy

82% Recall

SVM

Confusion Matrix	Predicted No Autism	Predicted Suspected Autism
Actual No Autism	TN 88	FP 4
Actual Suspect Autism	FN 9	TP 40

Conclusion

Question
Can we reduce the number of questions to achieve a comparable predictive performance to the original questionnaire?

Yes, Naive Bayes is the best model post optimisation for this data set.

- High Accuracy at 93%
- High recall rate of 92%

we can recommend removing questions 1 6 & 9 and still have a strong predictive model.

Future work:

- Further Analysis on a different set of questions to remove for better results.
- Perhaps removing 50/50 questions yield better result.

Thank You

Questions?